

AKHILESH KUMAR MISHRA

Software Development Engineer | Microservices · Distributed Systems · Full Stack

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PROFILE

Software Development Engineer building secure, high-scale distributed systems for national e-governance and FinTech platforms. Designs microservice architectures, transaction-critical APIs, and event-driven pipelines serving 2 million+ users and 500,000+ daily transactions. Polyglot across Java/Spring Boot, Python/FastAPI, and Node.js, with strong system design, database engineering, and observability practice. Led hiring and onboarding for a four-engineer team while standardizing Docker-based CI/CD delivery.

CORE TECHNOLOGY

Languages: Java, Python, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

Backend & Architecture: Spring Boot, Django, FastAPI, Node.js, Express.js, React.js, Next.js, microservices, REST APIs, WebSockets

Data & Messaging: MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, Celery, Hibernate/JPA, indexing & query tuning

DevOps & Observability: Docker, Git, GitHub Actions, CI/CD, Linux, Postman, Jira, Prometheus, Grafana

EXPERIENCE

Software Development Engineer — *CSC e-Governance Services India Ltd. (MeitY, Government of India)*

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic backend services into distributed microservices on **Spring Boot**, **FastAPI**, and **Node.js**, improving reliability for **2 million+ users**.
- Architected transaction-critical endpoints for AEPS, MATM, and UPI platforms, sustaining **500,000+ daily transactions** across payment rails.
- Secured platform APIs with **JWT** authentication and **RBAC**, adding device binding and OTP controls to close off unauthorized access paths.
- Designed and tuned **MySQL**, **PostgreSQL**, and **MongoDB** schemas using sharding, partitioning, and query optimization, cutting read/write latency by **45%**.
- Established event-driven communication with **Apache Kafka**, **Celery**, and **Redis**, lifting message processing throughput by **35%**.
- Instrumented services with **Prometheus** and **Grafana** and automated container delivery through **Docker** and **GitHub Actions**.
- Led hiring and onboarding for **4 engineers**, lifting team onboarding efficiency by **40%**.

Freelance Software Engineer — *Workforce Telemetry & Operations*

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Engineered a real-time workforce telemetry platform with a **React.js** front end and **FastAPI** backend, serving **1,000+ active field agents**.
- Built an asynchronous location ingestion pipeline on **FastAPI**, **Celery**, and **Redis**, processing **10,000+ concurrent GPS pings** at **99.9% uptime**.
- Delivered an interactive live map dashboard in **React.js** with Ant Design, cutting UI load times by **30%**.

SELECTED PROJECTS

Transaction Orchestration & Ledger Platform

Spring Boot / FastAPI / Kafka / PostgreSQL / Redis

Microservice platform orchestrating AEPS, MATM, and UPI settlement with a transactional wallet ledger, retry and reconciliation handling, and Kafka event fan-out.

MERN FinTech & Commerce Suite

React.js / Node.js / Express.js / MongoDB / JWT / Redux

Full-stack applications with JWT-secured authentication, role-based admin dashboards, and versioned REST APIs on React, Express, and MongoDB.

Real-Time Telemetry & Geofencing Engine

React.js / FastAPI / Celery / Redis / PostgreSQL

Workforce geolocation engine ingesting asynchronous GPS streams, rendering live map visualizations, and pushing geofence alerts over WebSockets.

Enterprise Authentication & Authorization Service

Spring Boot / Spring Security / JWT / MySQL

Reusable auth service providing registration, password hashing, JWT issuance and rotation, and RBAC enforcement across protected REST routes.

EDUCATION

Bachelor of Technology (B.Tech.) | **Kanpur Institute of Technology, Kanpur, UP**

2019 - 2023

CERTIFICATIONS

Data Structures & Algorithms Certification

Database Systems & SQL Certification

Advanced MERN Stack Developer Certification

IIT Kanpur Cyber Security Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.